RAJALAKSHMI ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

# Department of Cyber Security

## 2026-2027- Odd Semester

## III Year /V Sem

**CS23533-FOUNDATIONS OF ARTIFICIAL INTELLIGENCE**

## Review 1– 09.09.2026

**AI-Based Cyber Defence Allocation Using Game Theory**

**Puvanesu R (Reg. No: 241901085)**

# Introduction



Organizations face constant cyber threats but have only limited resources to secure every digital asset.

Domain: Artificial Intelligence — Adversarial Search and Game Theory applied to Cybersecurity Resource Allocation.

Deciding which assets to protect first is critical — the wrong choice can lead to major financial and reputational damage.

This project models Defender-vs-Attacker allocation as a two-player zero-sum game, solved using Minimax and Alpha-Beta search.

# Problem Statement



Security teams must decide which of many digital assets to protect with a limited budget, without a systematic way to compare options.

Existing approaches rely on manual judgment or static checklists, not optimal decision-making against an intelligent adversary.

A wrong allocation can leave high-value assets (e.g., payment gateways, customer databases) exposed to attackers who exploit the gaps.

An algorithmic, game-theoretic model is needed to compute the defense allocation that minimizes worst-case damage.

# Motivation



Cybersecurity is a core specialization for our department, and this project connects classical AI search to a real security decision problem.

Security budgets are always limited in practice; organizations need explainable, algorithm-driven prioritization rather than guesswork.

A transparent, repeatable method for allocating limited security resources to minimize potential damage.

Demonstrates how Unit III concepts (Game Theory, Adversarial Search) solve a genuine, socially relevant cybersecurity problem.

# Objectives



To model cybersecurity resource allocation as a two-player zero-sum adversarial game.

To implement the Minimax algorithm to compute the Defender's optimal allocation strategy.

To implement Alpha-Beta pruning and demonstrate its efficiency over plain Minimax.

To design a heuristic evaluation function so the search scales to larger, realistic scenarios.

To build an interactive interface where a user can play as Defender or Attacker against the AI.

# Literature Survey



| S.No. | Author & Year | Method / Technique | Key Findings | Limitations |
|---|---|---|---|---|
| 1 | Russell & Norvig, 2021 | Minimax & Alpha-Beta Search (AI: A Modern Approach) | Alpha-Beta pruning explores far fewer nodes than Minimax while returning the identical optimal move | Assumes a fully rational, worst-case opponent; does not model uncertain attacker behavior |
| 2 | Alpcan & Başar, 2010 | Game-Theoretic Modeling of Network Security | Game theory provides a principled framework for modeling attacker-defender interactions in cybersecurity | Real-world attacker behavior is often probabilistic, not perfectly rational, limiting pure-minimax accuracy |
| 3 | Roy et al., 2010 | Survey of Game Theory Applied to Network Security | Security resource allocation can be formulated as a Defender-Attacker game to guide investment decisions | Many proposed models are computationally expensive to scale to large real-world networks |

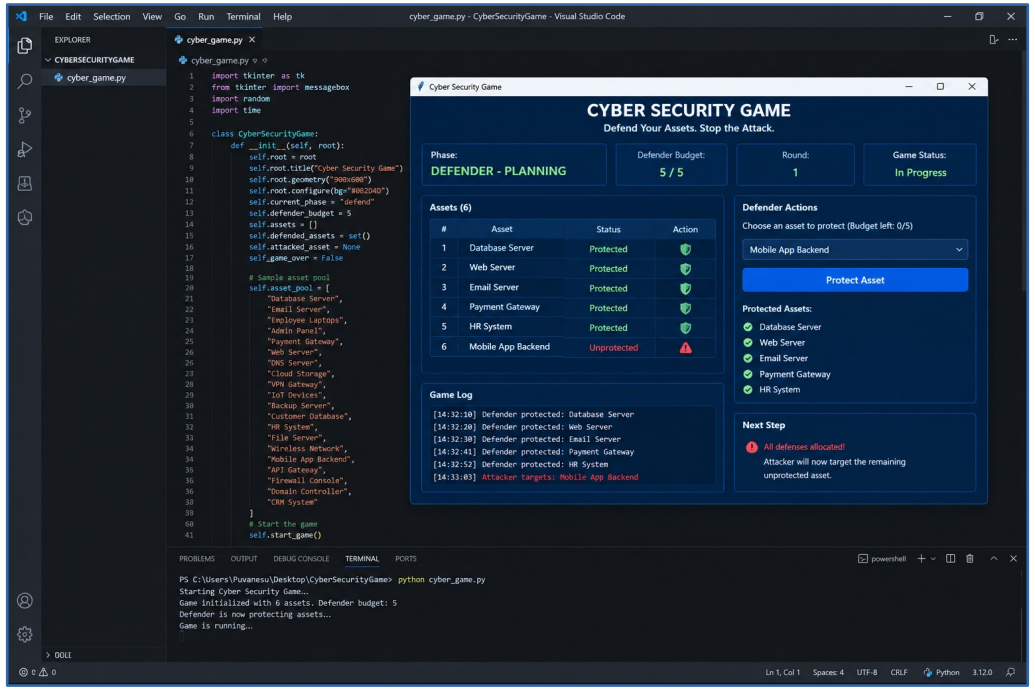
# Existing System

Manual risk assessment by security teams, using checklists and experience-based judgment.

Static vulnerability scanners and rule-based risk-scoring tools (e.g., CVSS-based prioritization).

Advantages: Simple to use; no specialized AI knowledge needed; widely adopted in industry.

Limitations: Ignores an intelligent, adaptive attacker; scores assets independently instead of jointly; inconsistent between assessors.



# Proposed System



Model Defender-vs-Attacker resource allocation as a two-player zero-sum game solved using Minimax.

Alpha-Beta pruning for efficiency; heuristic evaluation for scalability; Human-vs-AI or AI-vs-AI play modes.

Computes a provably optimal allocation against a worst-case attacker, instead of relying on manual judgment.

A working interactive system that recommends defense allocations and shows measurable efficiency gains from pruning.

**Assets & Budget → Game State → Minimax / Alpha-Beta Search → Optimal Allocation**

# System Architecture / Block Diagram






Data flows left to right through each stage of the system, from raw asset data to the final recommended allocation.

# Methodology



1. Define Assets & Scenario (value, vulnerability, defender budget)

2. Construct Initial Game State (Module 1)

3. Generate Legal Moves for Defender & Attacker

4. Apply Minimax / Alpha-Beta with Heuristic Evaluation (Module 2)

5. Compare Nodes Explored & Final Damage: Minimax vs Alpha-Beta

6. Display Optimal Allocation via Interactive Console/GUI

# Technologies Used



Programming Language: Python 3

Framework / Library: Tkinter (GUI); Python Standard Library only

Database: Not required (in-memory game state, no persistent storage)

Tools / IDE: Visual Studio Code

Hardware: Any standard PC/laptop (no GPU required)

# Module Description



**Module 1 – Game State & Environment**

**Module 2 – Adversarial Search Engine**

**Module 3 – Cybersecurity Scenario / Asset Pool**

**Module 4 – Interactive Play Controller**

**Module 5 – Graphical User Interface (Planned)**

# Preliminary / Expected Results



**Expected: Bar chart comparing nodes explored — Minimax vs Alpha-Beta pruning**

**Expected: Console output showing optimal Defender allocation & final damage score**

# Project Schedule



| Activity | Month 1 | Month 2 | Month 3 | Month 4 |
|---|---|---|---|---|
| Literature Survey | ✓ | ✓ | | |
| Problem Identification | ✓ | | | |
| System Design | | ✓ | ✓ | |
| Implementation | | | ✓ | ✓ |
| Testing & Evaluation | | | | ✓ |

# Current Progress



- ✓ Literature survey completed
- ✓ Problem identification completed
- ✓ Requirement analysis / methodology finalized
- ☐ Implementation completed
- ☐ Testing completed
- ☐ Final evaluation completed

# Future Work



Build the graphical user interface (Module 4) with full-screen, clickable Human-vs-AI gameplay.

Test across larger scenarios (10+ assets) to evaluate scalability of the heuristic evaluation function.

Compare Minimax vs Alpha-Beta performance (nodes explored, execution time) across varying game sizes.

Evaluate recommendations against manual/checklist-based allocation to demonstrate improvement.

Prepare the final report, user documentation, and presentation for the final review.

# Conclusion



This project models cybersecurity resource allocation as a two-player adversarial game, solved using Minimax, Alpha-Beta pruning, and heuristic evaluation.

Combines three classical AI search techniques into one system and applies them to a genuine, socially relevant cybersecurity decision problem.

The completed system will give organizations with limited security budgets a transparent, explainable way to prioritize defenses against a rational adversary.

# References



[1] S. Russell and P. Norvig, “Artificial Intelligence: A Modern Approach,” 4th ed., Pearson, 2021.

[2] T. Alpcan and T. Başar, “Network Security: A Decision and Game-Theoretic Approach,” Cambridge University Press, 2010.

[3] S. Roy et al., “A Survey of Game Theory as Applied to Network Security,” Proc. 43rd Hawaii Int. Conf. on System Sciences (HICSS), 2010.

# THANK YOU

Questions & Suggestions